
Diffusion scores for Markov data: exact inference on the chain and estimation of the transition kernel

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Internal note, work in progress. Circulated for discussion. The inference and estimation results are settled and reproducible; the generative comparison of §8 is complete for one innovation family with four more in progress, and its interpretation is the main open question. Section 9 states what is not yet supported. Open items for the advisors are listed separately in QUESTIONS_FOR_ADVISORS.md.

Abstract

Score-based generative models are trained by denoising regression, and the function they approximate — the posterior mean of the clean signal given its noised version — is unavailable in closed form for essentially every distribution of interest. This makes it hard to separate the error a network makes from the difficulty of the object it approximates. We study a setting where the posterior mean is computable and non-trivial: a first-order Markov chain on $\mathbb{R}^n$ with continuous, non-Gaussian transitions. Coordinatewise Ornstein–Uhlenbeck noising multiplies the prior by unary factors, so the posterior remains a chain, and sum-product inference on it is exact at the level of functional messages. We use this to obtain a controlled benchmark, and then remove the assumption that the transition kernel is known: it is estimated from noised sequences alone by maximum marginal likelihood, with an exact expectation step and a gradient supplied by Fisher’s identity, so the inference recursion is never differentiated through. On a Laplace chain the estimator attains a lower denoising error than trained neural denoisers at every data budget we tried, by roughly an order of magnitude against a fully connected network and a factor of four against a locality-respecting one. We then compare the same estimators through reverse diffusion, and the ranking does not carry over: the structured estimator remains the most accurate pointwise while the convolutional network reproduces the innovation law more faithfully. We report this dissociation as the principal empirical finding rather than resolving it, and we identify the information-theoretic asymmetry that plausibly explains it.

1 Introduction

Generative diffusion generates data $a \in \mathbb{R}^n$ from a distribution P_0 known only through samples, by noising and then reversing [Sohl-Dickstein et al., 2015, Ho et al., 2020, Song et al., 2021]. In the Ornstein–Uhlenbeck convention we follow throughout [Mézard, 2025], the forward process is

$$\frac{dx}{dt} = -x + \sqrt{2}\eta(t), \quad \text{so} \quad x(t) = e^{-t}a + \sqrt{\Delta_t}z(t), \quad \Delta_t = 1 - e^{-2t}, \quad (1)$$

with z standard Gaussian. Reversal is driven by the score $S(x, t) = \nabla_x \log P_t(x)$, and the central quantity is the posterior mean $\langle a \rangle_{x,t}$ of the clean signal.

Three distinct score objects. Confusing them is the source of a common misstatement, so we fix them at the outset.

The *population score* is $\nabla_x \log P_t(x)$ for the true P_0 . The *empirical score* replaces P_0 by the empirical measure $\hat{P}_0 = M^{-1} \sum_{\mu} \delta_{a^\mu}$ of the training set; after noising this is a Gaussian mixture centred on the training points. The *learned score* is what a network with finite capacity, finite data and a particular optimiser actually produces.

The population denoising objective $\mathbb{E} \|f(x, t) - a\|^2$ is minimised over all measurable f by the population posterior mean. The *empirical* objective, which is what is computed in practice, is minimised by the *empirical* posterior mean. Since exact reversal under the empirical score returns the empirical distribution, a model that attained the unrestricted optimum of its own training objective would reproduce its training set. That trained models generalise is therefore a statement about the restrictions — architecture, capacity, optimisation, regularisation — and not about the objective [Biroli et al., 2024, Kamb and Ganguli, 2024]. We do not claim this is a complete account of neural generalisation; we note it because it motivates wanting settings where the population score is available for comparison.

Why a Markov chain. Distributions with tractable population scores are scarce, which is why the analytical literature leans on Gaussians and Gaussian mixtures. Those are poor probes of structure: the score is linear, so any question about how a model exploits correlation has a degenerate answer. Hierarchical and discrete graphical models supply further tractable families [Sclocchi et al., 2025, Garnier-Brun et al., 2024, Mei and Wu, 2024], and the model here is a continuous, non-Gaussian member of the same class.

A first-order Markov chain gives a genuinely nonlinear score while remaining exactly computable, for a structural reason: coordinatewise noising contributes one unary factor per site, which reweights node potentials without creating edges between latent variables, so the posterior factor graph is the same chain as the prior’s.

Questions. We ask: (i) what does exact inference on the chain buy over the Gaussian approximation, and where in the noise schedule does it matter; (ii) can the transition kernel be estimated from noised data alone, and at what cost; (iii) how does the resulting estimator compare with trained denoisers, both pointwise and after reverse diffusion.

Contributions.

- An exact denoising benchmark for a continuous non-Gaussian correlated model (§3), with the discretisation error separated from the model error and measured (§4).
- An identification of what the single-Gaussian message closure computes: exactly the LMMSE estimator of the covariance-matched Gaussian model (§5).
- Estimation of the unknown kernel from noised sequences, with an exact expectation step and no differentiation through the recursion (§6).
- A denoising comparison against neural baselines (§7) and a generative comparison (§8) which do not agree, together with the information asymmetry that plausibly explains the disagreement.

2 Setting

Notation. We write n for the chain length, which is also the ambient dimension, and M for the number of observed sequences. (The source we follow for diffusion notation uses n for the number of samples and d for the dimension; we index by the chain because every graphical-model statement below is indexed by sites.) Sequences are $a = (a_1, \dots, a_n)$, and P_0 factorises as

$$P_0(a) = \mu(a_1) \prod_{i=2}^n W(a_i | a_{i-1}), \quad W(a' | a) = \varphi(a' - \rho a), \quad (2)$$

with φ an innovation density of mean zero and variance q , and $|\rho| < 1$. Choosing $q = 1 - \rho^2$ makes the chain stationary with unit marginal variance and $\text{Cov}(a_i, a_j) = \rho^{|i-j|}$ *independently of the shape*

of φ . This is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations can be arranged to share a covariance exactly and differ only beyond second moments.

The score and the posterior mean. From (1), P_t is the Gaussian convolution

$$P_t(x) = \int da P_0(a) \frac{1}{(2\pi\Delta_t)^{n/2}} \exp\left[-\frac{\|x - e^{-t}a\|^2}{2\Delta_t}\right]. \quad (3)$$

Differentiating under the integral,

$$\nabla_x P_t(x) = \int da P_0(a) \frac{e^{-t}a - x}{\Delta_t} \frac{e^{-\|x - e^{-t}a\|^2/2\Delta_t}}{(2\pi\Delta_t)^{n/2}},$$

and dividing by $P_t(x)$, the Gaussian factor over $P_t(x)$ is exactly the posterior density $P(a | x, t)$, so

$$S(x, t) = \nabla_x \log P_t(x) = -\frac{x - e^{-t}\langle a \rangle_{x,t}}{\Delta_t}, \quad \langle a \rangle_{x,t} = \int da a P(a | x, t). \quad (4)$$

Differentiation under the integral is justified whenever P_0 has a finite first moment, by dominated convergence with the Gaussian kernel and its derivative. Equation (4) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem.

The Bayes-optimal denoiser. For squared loss, and for any u ,

$$\mathbb{E}[\|a - u\|^2 | x] = \mathbb{E}[\|a - \langle a \rangle_{x,t}\|^2 | x] + \|u - \langle a \rangle_{x,t}\|^2, \quad (5)$$

so $\langle a \rangle_{x,t}$ minimises the conditional risk, with an irreducible floor given by the first term. Optimality is with respect to squared error specifically; a different loss selects a different functional of the posterior.

3 Exact inference on the noised chain

Conditioning (2) on x gives

$$P(a | x, t) \propto \mu(a_1) \prod_{i=2}^n W(a_i | a_{i-1}) \prod_{i=1}^n \ell_t(x_i | a_i), \quad \ell_t(x_i | a_i) \propto \exp\left[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right]. \quad (6)$$

The likelihood factors are *unary*: because the forward noise is independent across coordinates, each observation touches one latent variable. They therefore reweight node potentials and add no edges, and the posterior factor graph is the prior's chain (Figure 1). A chain is a tree, so sum-product is exact [Pearl, 1988, Mézard and Montanari, 2009]: with

$$\begin{aligned} \vec{m}_i(a_i) &\propto \int da_{i-1} W(a_i | a_{i-1}) \ell_t(x_{i-1} | a_{i-1}) \vec{m}_{i-1}(a_{i-1}), \\ \overleftarrow{m}_i(a_i) &\propto \int da_{i+1} W(a_{i+1} | a_i) \ell_t(x_{i+1} | a_{i+1}) \overleftarrow{m}_{i+1}(a_{i+1}), \end{aligned} \quad (7)$$

the single-site marginal is $P(a_i | x, t) \propto \vec{m}_i(a_i) \ell_t(x_i | a_i) \overleftarrow{m}_i(a_i)$ and the pairwise marginal on edge $(i-1, i)$ is $\propto \vec{m}_{i-1}(a_{i-1}) \ell_t(x_{i-1} | a_{i-1}) W(a_i | a_{i-1}) \ell_t(x_i | a_i) \overleftarrow{m}_i(a_i)$. In (7) the local likelihood is absorbed *before* propagation, so $\ell_t(x_i | a_i)$ appears exactly once in each marginal; this convention is fixed throughout.

Remark 1 (What “exact” means here). *Three statements must be kept apart. (i) Sum-product on the posterior (6) is exact at the level of functional messages, as a consequence of the graph being a tree. (ii) Messages are functions on $\mathbb{R}$, so an implementation must represent them; we use a truncated uniform grid with composite trapezoidal weights, which defines a finite-state model for which the recursion is exact up to floating-point arithmetic. (iii) That finite-state model approximates the continuous one, with an error that must be measured rather than assumed. We write “exact tree inference” for (i) and “exact for the discretised model” for (ii), and never abbreviate either to “exact”.*

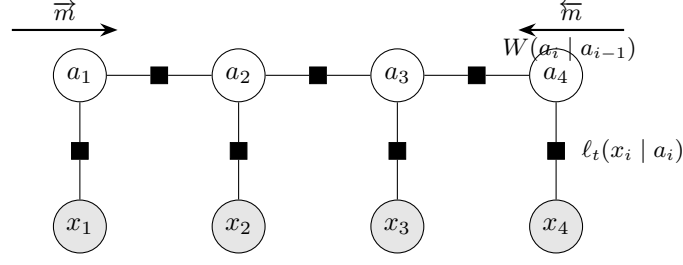


Figure 1: Factor graph of the posterior (6). Latent variables a_i (open) carry binary transition factors along the chain; each observation x_i (shaded) enters through a *unary* factor ℓ_t attached to a single latent variable. Conditioning therefore reweights node potentials without creating edges among the a_i , and the posterior graph remains a chain. Sum-product with the messages (7) is exact on it.

4 Numerical control

Two error sources are separated by construction. *Truncation*: probability mass outside the grid $[-A, A]$ is discarded, monitored by the mass in the boundary cells, which stays below 2×10^{-5} across all reported runs at $A = 8$. *Quadrature*: the trapezoidal rule is second order in the spacing for integrands with a bounded second derivative. For the Gaussian kernel the integrand is smooth and convergence is fast; **for the Laplace kernel it is not**, since $\varphi(e) \propto e^{-|e|/b}$ has a discontinuous derivative at the origin, and we therefore make no claim of spectral accuracy for that family.

Against the closed-form Gaussian posterior mean, the discretised recursion agrees to 9.2×10^{-15} relative at $K = 401$, $A = 8$. That number measures the discretisation of the Gaussian model only. It is not comparable with the independent Monte Carlo check below, which measures something else: on the Laplace chain, where no closed form exists, importance sampling with 6×10^6 samples agrees with the discretised posterior mean to 1.1×10^{-3} relative — a figure limited by the Monte Carlo standard error, not by the grid. Both are reported because they validate different things; neither is the accuracy of the other.

5 Gaussian closure and the LMMSE estimator

A standard approximation propagates only the first two moments of each message. Under the transition $a_i = \rho a_{i-1} + \varepsilon_i$, a message with mean m and variance v maps to one with

$$m \mapsto \rho m, \quad v \mapsto \rho^2 v + q, \quad (8)$$

and a Gaussian message times a Gaussian likelihood is Gaussian. The propagation therefore uses only the mean and variance of φ and discards everything else about its shape.

Proposition 1. *For the chain (2) with linear transitions and Gaussian unary likelihoods, moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\hat{a}_{\text{LMMSE}} = \text{Cov}(a, x) \text{Cov}(x)^{-1} x = e^{-t} \Sigma_0 (e^{-2t} \Sigma_0 + \Delta_t I)^{-1} x, \quad (9)$$

for every innovation law with mean zero and variance q .

The proof is the observation that (8) depends on φ only through $(0, q)$, so the whole recursion is a function of Σ_0 ; the resulting estimator is linear in x and matches the two moments, which characterises LMMSE. We verify it numerically to 6×10^{-16} across five innovation families.

Remark 2. *Proposition 1 is a statement about this closure — single-Gaussian moment projection with linear transitions — and not about Gaussian approximations in general. Within its scope it has a consequence worth stating: what one might call the message-representation error and the model error coincide, because projecting the messages and replacing the model by its covariance-matched Gaussian give the same estimator. The two separate again for richer message families.*

The gap between (9) and exact inference locates where non-Gaussianity matters. It is largest at *small* t and decays monotonically: on a Laplace chain at $\rho = 0.85$, the relative deviation of the closure’s score from the exact one falls from 0.20 at $t = 0.08$ to 9.2×10^{-5} at $t = 2.4$. Large t Gaussianises the posterior, so the shape of φ survives only where the likelihood is weak.

6 Estimating the transition kernel

We now treat W as unknown and observe only noised sequences x^μ , $\mu = 1, \dots, M$, possibly at several noise levels. The initial law $\mu(a_1)$ is held fixed (see §9). Write θ for the kernel parameters and $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(x^\mu)$.

Parametrisation. We use a mixture-innovation kernel

$$W_\theta(a' | a) = \sum_{c=1}^C \pi_c \mathcal{N}(a' - \rho a - \nu_c, s_c^2), \quad (10)$$

which keeps the linear autoregression and leaves the innovation law essentially free. At $C = 4$ this has ρ , four weights π_c , four locations ν_c and four scales s_c^2 : thirteen numbers subject to $\sum_c \pi_c = 1$, hence **twelve free parameters**. It is never given the density it is asked to recover.

Exact expectation step. Many latent-variable models require an approximate or variational E-step; this one does not, because the posterior is a tree (§3) and sum-product returns exact single-site *and pairwise* marginals. The complete-data log-likelihood is a sum over edges, so the auxiliary function $\mathcal{Q}(\theta | \theta^{(r)})$ depends on the data only through

$$\Xi_{kl} = \sum_{\mu=1}^M \sum_{i=2}^n P(a_{i-1} = u_k, a_i = u_l | x^\mu), \quad (11)$$

the expected transition mass between grid points u_k and u_l — the continuum analogue of the expected transition counts of Baum–Welch [Baum et al., 1970]. Sufficiency of the single matrix (11) requires a *homogeneous* kernel shared across sites and sequences, a fixed grid, and an initial law excluded from θ ; it fails if any of these is relaxed. Given those, the maximisation step never revisits the data and costs $O(PK^2)$ for P parameters, independently of M ; observations at different noise levels add into one Ξ . Forming Ξ costs $O(MnK^2)$ per iteration and $O(K^2)$ storage, so “one matrix” is a statement about sufficiency, not about inference being cheap.

Gradient without differentiating the recursion. Fisher’s identity [Fisher, 1925, Cappé et al., 2005] gives

$$\nabla_\theta \log P_\theta(x) = \mathbb{E}_\theta[\nabla_\theta \log P_\theta(a, x) | x] = \langle \Xi(\theta), \nabla_\theta \log W_\theta \rangle, \quad (12)$$

the exact gradient of the *marginal* log-likelihood from one sum-product pass. Sum-product is not a computation graph to backpropagate through; it supplies the conditional expectation. We verify (12) against finite differences of the exact evidence to $\sim 10^{-9}$.

Equation (12) is not EM. It licenses *gradient ascent* on $\mathcal{L}$, which is the procedure described in the guidance we were given, whereas EM maximises $\mathcal{Q}$ at each step. Both are implemented and they do not dominate one another: on the smooth Gaussian kernel gradient ascent reaches EM’s optimum to 2×10^{-9} nats while requiring a tuned step size, and on the Laplace kernel it attains a higher likelihood than EM, because there the exact maximiser of $\mathcal{Q}$ is a lattice artefact of the discretisation. Results below use EM with the exact maximisation step unless stated.

Identifiability. Two claims must be separated. Gaussian convolution at finite t is injective on distributions — the characteristic function acquires a factor $e^{-\Delta_t \|\xi\|^2/2}$, which has no zeros, and $e^{-t} > 0$ — so P_t determines P_0 . *Kernel* identifiability does not follow: it additionally requires the initial law to be fixed or separately identifiable, the transition family to be identifiable as a parametric family, and the mixture (10) to be treated modulo label permutation. We assume these rather than prove them.

What noising does destroy is *information*, and unevenly. Measured per sequence from exact single-chain passes, the Fisher information for the innovation scale falls by a factor 142 between $t = 0.05$ and $t = 1.6$, while that for the correlation falls by 26. The higher-order structure of φ is thus far more fragile under the channel than its second-order structure — a fact we return to in §8.

Table 1: Relative denoising error against exact inference under the true kernel, averaged over the noise schedule. Mean over six training seeds, $\pm$ one standard error. The seeds vary initialisation at fixed data, so these intervals measure optimisation variance and not sampling variance.

sequences M	network (25,248 params)	EM-BP (12 free)	ratio
32	0.6363 ± 0.0043	0.0651 ± 0.0031	9.8
128	0.5117 ± 0.0025	0.0333 ± 0.0035	15.4
512	0.2813 ± 0.0013	0.0211 ± 0.0022	13.3
1024	0.2154 ± 0.0009	0.0182 ± 0.0021	11.8
2048	0.1756 ± 0.0009	0.0142 ± 0.0009	12.4
4096	0.1643 ± 0.0009	0.0124 ± 0.0005	13.2

7 Denoising accuracy

Protocol. Laplace chain, $n = 32$, $\rho = 0.85$. Estimators are scored against exact-BP denoising under the true kernel on held-out sequences, at five noise levels. Neural baselines are trained by denoising score matching in both standard parametrisations (predicting the noise, and predicting the clean signal); we report the better of the two at each noise level, since they win in different regimes and reporting one alone would mislead. Full architectural and optimisation details are in Appendix A.

Table 1 shows a ratio between 10 and 15 across seven doublings of the data. The comparison at equal budget is the meaningful one. We note that the network at $M = 4096$ did not reach the error the estimator attains at $M = 32$; **we did not observe a crossing**, so we do not quantify a data-equivalence factor, and the extrapolation that would be required to do so spans several decades.

Controls. Capacity is not the explanation: over 24 configurations from 3.2k to 297k parameters the best network reaches 0.208 against the estimator’s 0.022 at $M = 1024$, and error degrades for the largest models. The baseline is competent: trained on a Gaussian chain where the optimal denoiser is linear and closed-form, given four times the data and twice the steps, it reaches 0.099.

A locality-respecting baseline. A fully connected network carries no sequential prior. Against a weight-shared window predictor — a one-dimensional convolution — the margin narrows substantially, from roughly 10 to between 2 and 4 over $M \in \{128, 512, 2048\}$. Roughly 60% of the fully connected network’s deficit was therefore its architecture rather than the estimator. The margin against the convolution increases over the three budgets we ran (2.10 ± 0.39 , 3.27 ± 0.32 , 4.11 ± 0.32); three points do not establish a trend, and we report them as measured. **These particular numbers used a receptive field selected on the evaluation set** and are exploratory pending a rerun with a validation split; the bias favours the baseline, so the margin quoted is if anything conservative.

8 Generation

Denoising accuracy is not what a generative model is judged on. Errors enter the reverse integration at every step and accumulate unevenly across the schedule, so a pointwise ranking need not survive. We therefore integrate the reverse process $dx/d\tau = x + 2S(x, \tau) + \eta$ under each estimator, with common random numbers across estimators.

Comparing like with like. Integration must stop at $t_{\min} > 0$, since S diverges as $\Delta_t \rightarrow 0$. What each estimator produces is therefore a sample of $P_{t_{\min}}$, not of P_0 , and comparing it against clean data measures the stopping point rather than the score: independent Gaussian noise of variance v on an innovation of variance q multiplies the excess kurtosis by $(q/(q+v))^2$. We instead draw a reference from P_0 and noise it forward to the same $t_{\min}$, so both sides are samples of the same law and the target moments follow in closed form. Run with the *true* kernel, the sampler reproduces that target at every budget (1.897–1.914 against 1.9098), which validates the integrator and the statistics before any estimator is compared.

Table 2 is the paper’s main empirical finding, and it is a dissociation. The structured estimator is an order of magnitude more accurate pointwise and three times better on second-order structure, and

Table 2: Generated-sequence statistics at $M = 2048$, Laplace chain, median over four seeds, against closed-form targets for $P_{t_{\min}}$. The denoising column repeats the ranking of §7; the distributional columns do not agree with it.

	EM-BP	local CNN	global MLP	target
denoising MSE against exact	0.00052	0.00525	0.01330	0
innovation excess kurtosis	0.897	1.357	0.206	1.910
$D_{\text{KL}}(\text{true} \parallel \text{generated})$	0.0055	0.0028	0.0165	0
worst covariance-lag error	0.0292	0.0973	0.0518	0

yet the convolutional network reproduces the *innovation law* more faithfully, by both kurtosis and divergence. The fully connected network shows the pattern most sharply: over the same range its denoising error improves twelvefold while its generated kurtosis degrades from 0.850 to 0.206. It fits the bulk of the posterior and loses the tail.

A plausible mechanism, not yet established. Both the marginal likelihood that EM maximises and the squared-error denoising loss are dominated by the bulk of the distribution. The tail of φ carries little probability mass and therefore little weight in either objective, while contributing disproportionately to the shape statistics that reverse diffusion exposes. This is the same asymmetry measured in §6: the channel destroys innovation-shape information 142-fold against 26-fold for the correlation, and the estimator sees only noised data. Whether the deficit is a capacity limit of the parametrisation (10) or an information limit of the observation model is decided by varying C , which is in progress and not reported here.

9 Limitations and scope

The model assumes exact first-order Markov structure; under a non-Markov prior the estimator is misspecified in a way more data does not repair. The transition family (10) may additionally be misspecified, and §8 suggests it is, for the tails. The initial law is fixed rather than estimated. Sum-product is exact on the posterior chain but the implementation is a discretisation whose error, for the Laplace family, is second order rather than spectral.

Inference costs $O(nK^2)$ per sequence per evaluation against a few matrix products for a network, measured here at two to three hundred times slower; since reverse diffusion calls the denoiser at every step, this is the practical weak point. Training is also slower, and the estimator’s cost grows linearly in M while the network’s is fixed by its step count, so the advantage is in data and not in wall-clock. The low parameter count is not free: it is bought with a structural assumption, and the advantage narrows as the parametric dimension grows — on a discrete-alphabet variant it falls from 15 to 4.4 as the estimated dimension goes from 6 to 132 parameters.

Error bars over training seeds measure optimisation variance at fixed data. The generative comparison is complete for one innovation family. Comparisons are against the two architectures described and do not license a claim about neural denoisers in general.

10 Conclusion

When a sequential distribution is known to have low-dimensional Markov structure, learning a diffusion score reduces to latent-variable inference and estimation of a compact transition model. Sum-product on the noised chain supplies an exact non-Gaussian benchmark, and estimating the kernel from noised data alone gives a denoiser that is substantially more accurate at small sample sizes than trained networks, at the cost of stronger assumptions and much more expensive inference. The comparison after reverse diffusion does not reproduce that ranking, and we regard understanding the dissociation — rather than the pointwise margin itself — as the useful direction.

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